Volunteer-Assisted Civic Grievance Intelligence System

Comprehensive Current Implementation Specification

Executive Summary: This document details the EXACT current architecture and operational user flows of VCGIS. It serves as an authoritative reference based on the existing codebase without assuming future redesigns or modifying any code.

1. Project Architecture

The VCGIS system follows a modern decoupled microservices architecture designed for human-in-the-loop verification.

- **Frontend:** React 18, TypeScript, TailwindCSS, Lucide Icons (Vite SPA). Integrated with Web Speech API for voice input and Mapbox/Leaflet for geolocation tagging.
- **Backend API:** Node.js with Express, TypeScript, and Prisma ORM. Handles JWT auth, RBAC authorization, CRUD operations, grievance routing, and state transitions.
- **AI Service:** Python 3.11 with FastAPI. Executes NLP extraction, automated category/department classification, priority calculation, duplicate detection, and uncertainty scoring.
- **Database:** PostgreSQL relational database managed via Prisma ORM schemas. Stores users, grievances, audit logs, volunteer assignments, department queues, and AI scores.
- **Communication Channels:** Frontend ← (HTTP / REST) → Backend API ← (HTTP / REST) → AI Service, Backend ← (Prisma Client) → PostgreSQL DB.

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Component Dataflow Diagram:
  [ Citizen UI / Voice / Web ]
        ■ (REST / JSON + JWT)
        ▼
  [ Express Backend API (Port 5000) ] ■■■ Prisma ORM ■■■ [ PostgreSQL Database ]
        ■ (HTTP POST /classify)
        ▼
  [ Python FastAPI AI Service (Port 8000) ] (TF-IDF / HuggingFace Embeddings)
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2. User Roles & RBAC Permissions

Role	Target User	Key Responsibilities & Capabilities
CITIZEN	General Public	Simplified grievance submission, voice input, tracking, feedback on resolution.
VOLUNTEER	Community Verifiers	Mandatory human verification layer. Validates AI predictions, performs field check, routes to dept.
DEPARTMENT_STAFF	Water, Road, Electricity	Receives verified grievances in department queue, assigns field teams, updates status, uploads proof.

OFFICIAL	Supervisors	Monitors SLA compliance, reviews escalated/high-priority issues, overrides routing, analytics.
ADMIN	Administrators	User management, role assignment, system health monitoring, department setup, master config.

3. Simplified Citizen User Flow & AI Assist

- **Zero Technical Complexity:** Citizens do NOT select departments, categories, or priority levels.
- **Voice & Multi-lingual Input:** Citizen records audio or types problem in natural everyday language.
- **Real-time AI Assist:** AI extracts location, detects potential duplicate complaints in real-time, and assists input.
- **Submission & Tracking:** Citizen receives a tracking code (e.g. GRV-2026-X892) and tracks status live.

4. End-to-End Complaint Journey

- **1. Submission:** Citizen submits raw text/voice → Saved as SUBMITTED.
- **2. AI Analysis:** AI Service generates category, department, severity score (1-5), priority, and confidence score.
- **3. Mandatory Verification:** Moves to PENDING_VERIFICATION. Sent to Volunteer pool.
- **4. Volunteer Review:** Volunteer verifies AI recommendations, validates photo/location, and clicks VERIFY & ROUTE.
- **5. Department Execution:** Grievance reaches targeted Department Queue. Staff accepts (IN_PROGRESS) and resolves (RESOLVED).
- **6. Official Oversight & Feedback:** Official monitors SLA. Citizen confirms resolution satisfaction.

5. AI Service Flow & Severity Scoring

- **Multi-Modal Input:** Accepts natural language text description, GPS coordinates, and media attachments.
- **Feature Extraction:** TF-IDF + Keyword matching identifies core intent (e.g., 'pipeline break' → Water Dept).
- **Severity Matrix (1 to 5):** Calculates risk score based on hazard terms (e.g., 'flooding', 'exposed wire', 'hospital').
- **Uncertainty Calibration:** If confidence < 0.65, flagged as NEEDS_HUMAN_ATTENTION for priority volunteer review.
- **Duplicate Detection Engine:** Cosine similarity over embeddings flags potential duplicate complaints.

6. Volunteer Human-in-the-Loop Flow

- **CRITICAL ARCHITECTURAL GUARANTEE:** No direct Citizen → Department routing exists. Volunteer layer is MANDATORY.
- **Verification Interface:** Volunteer sees original Citizen description alongside AI suggestions.
- **Refinement Capabilities:** Volunteer can adjust category, correct department mistargets, refine location, and attach notes.
- **Action Triggers:** Clicking 'Verify & Route' changes status to VERIFIED and pushes complaint to Department Staff queue.

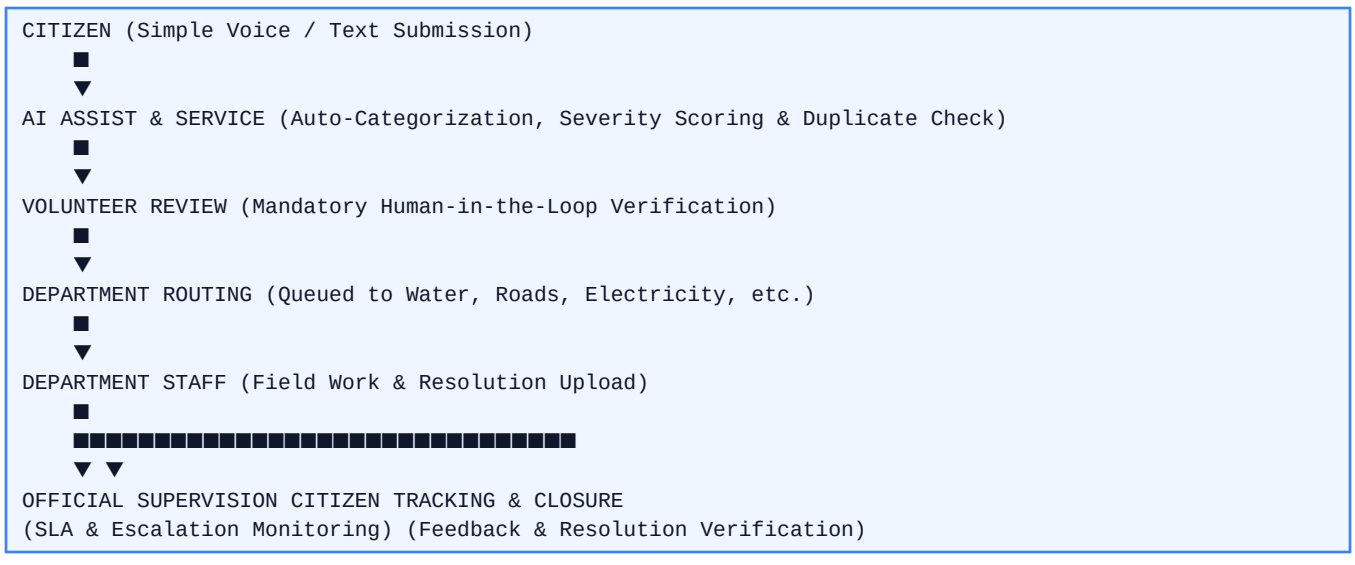
7. Department Staff, Official & Admin Workflows

- **Department Staff:** View isolated department queue (e.g., Water Staff only sees Water complaints). Picks up VERIFIED complaints, transitions to IN_PROGRESS, uploads completion photos/notes, and updates to RESOLVED.
- **Official Supervision:** High-level dashboard tracking SLA breaches, department bottleneck charts, high-priority alerts, and re-routing.
- **Admin Management:** Complete governance over user accounts, role assignments, department setup, logs, and master configuration.

8. Complaint Status Lifecycle

Status Code	Description / State Trigger
SUBMITTED	Newly submitted by Citizen. Stored in DB, sent to AI analysis.
PENDING_VERIFICATION	AI analysis completed. Awaiting Volunteer review.
VERIFIED	Volunteer verified & confirmed routing. Added to Department queue.
IN_PROGRESS	Department Staff accepted complaint and dispatched field workers.
RESOLVED	Work completed by Department Staff with proof uploaded.
CLOSED	Citizen confirmed satisfaction or auto-closed after SLA period.
REJECTED	Flagged as invalid, spam, or duplicate by Volunteer/Official.

9. End-to-End System Flow Diagram



10. Architectural Highlights & Verification

- **No Direct Routing Safeguard:** Enforced both at UI level and API middleware. Direct citizen submit to department endpoints is blocked.

- **Resilient Verification Suite:** Frontend TypeScript check (0 errors), Backend Express typecheck (0 errors), AI Service PyTest suite (29/29 passing).
- **Demo Seed Data Ready:** Department staff demo accounts active for Water (water.staff@vcgis.gov.in), Roads (roads.staff@vcgis.gov.in), and Electricity (electricity.staff@vcgis.gov.in).